O-S-605d
February 23, 1967

SUPERSEDING
Int. Fed. Spec. O-S-00605c (Army-GL)
December 7, 1965 and
Fed. Spec. O-S-605b
September 1, 1960

FEDERAL SPECIFICATION

SODIUM SILICATE SOLUTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers one type of sodium silicate solution (see 6.1).

1.2 **Classification.** The sodium silicate solutions covered by this specification shall be of one grade and shall be of the following classes as specified (see 6.2):

Class 1—41.5° Baumé.
Class 2—34.0° Baumé.
Class 3—52.0° Baumé.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-C-300—Chemicals, Liquid; Packaging and Packing of.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Physical and chemical.** The sodium silicate solutions shall conform to the requirements specified in table I when tested as specified in 4.3.2.

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TABLE I. *Physical and chemical requirements*

	Class 1	Class 2	Class 3
Bubble viscosity at 25° C., equivalent to a tube form, (approximate stokes)	C through T (0.85 to 5.50)	E through N (1.25 to 3.40)	T through Z (5.50 to 22.70)
Specific gravity at 20°/20° C., Baumé	40.5° to 42.5°	33.5° to 35.5°	51.5° to 53.5°
Sodium oxide (Na_2O), percent by weight	8.7 to 9.4	6.3 to 7.0	13.6 to 14.8
Weight ratio of silicon dioxide to sodium oxide (Parts SiO_2 to 1 part Na_2O)	3.17 to 3.27	3.70 to 3.80	2.35 to 2.45

3.2 **Labeling.** Each unit container of sodium silicate solution shall be clearly and legibly lithographed or printed on the body in water-and-compound resistant black ink as follows:

SODIUM SILICATE SOLUTION
CLASS (see 1.2)
(Stock Number)
CAUTION: Avoid contact with skin and eyes.
In case of contact, immediately flush skin or eyes with plenty of water for 15 minutes; FOR EYES GET MEDICAL ATTENTION

3.3 **Workmanship.** The sodium silicate solutions shall be colorless, clean, and homogeneous.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Certificates of compliance.* Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 **Inspection.** Sampling for inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated.

4.2.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 **Inspection of the end item.**

4.3.1 *Examination of the end item.* Th end item shall be examined for the defects in 4.3.1.1 at inspection level S-2 and acceptable quality level (AQL) of 2.5, expressed in defects per 100 units. A lot shall consist of all unit containers of the same capacity containing a sodium silicate solution produced under essentially the same manufacturing conditions, with no change in materials and provided the operation is continuous (see 5.1). In the event the process is a batch operation, each batch shall constitute a lot.

4.3.1.1 *Examination of workmanship.* An examination shall be made to determine that workmanship complies with the requirements of this specification. The sample unit for this examination shall be one unit container.

Examine	Defect
Workmanship	Presence of foreign matter.
	Not colorless.
	Not uniform.

4.3.1.2 *Examination of preparation for delivery.* The packaging, packing, and marking requirements of this specification

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shall be examined in accordance with the provisions of PPP-C-300, except that the inspection level shall be S-2 and the AQL shall be 2.5, expressed in defects per hundred units.

4.3.2 *Testing of the end item.* The methods of testing specified in Fed. Test Method Std. No. 141 and table II shall be followed for each lot. The lot size shall be expressed in gallons of sodium silicate solution. The sample unit for testing shall be a one-pint composite selected in equal portions from samples selected at random throughout the lot. The samples shall be selected in the frequency listed below and thoroughly mixed to form the composite. The composite shall be placed in a clean, dry container and sealed to protect against moisture. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite sample fails to meet any of the test requirements specified.

Lot size (gallons)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.4 **Test methods.** Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant.

4.4.1 *Specific gravity.* The specific gravity of the sodium silicate solutions shall be determined by use of a hydrometer, as specified in method 4183 of Fed. Test Method Std. No. 141, except a Baumé hydrometer shall be used and the specific gravity shall be determined at 20°/20°C.

4.4.2 *Sodium oxide.* Weigh 30 milliliters (ml.) of the sodium silicate solution to the nearest milligram in a weighing bottle; wash into a 1000-ml. volumetric flask, and dilute to volume with water. Transfer a 25-ml. aliquot portion to a 500-ml. beaker and

TABLE II. *Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to	Number of determinations per unit	Results reported as	
	Requirement	Test method	Comp. samp.		Pass or fail	Numerically to nearest
Bubble viscosity at 25° C.	Table I	4271[1] Procedure B	X	1	X	
Specific gravity at 20°/20° C.	Table I	4.4.1	X	2		Average of 2 determinations to nearest 0.1° Baumé
Sodium oxide (Na_2O), percent	Table I	4.4.2	X	2		Average of 2 determinations to nearest 0.1 percent
Weight ratio of silicon dioxide to sodium oxide (SiO_2/Na_2O)	Table I	[2]				

[1] Test shall be performed in accordance with Fed. Test Method Std. No. 141.

[2] Certificate of compliance shall be furnished the contracting officer or his authorized representative certifying that the product conforms to this requirement.

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add 75-ml. of water and 2 to 3 drops of methyl orange indicator solution. Titrate with 0.1 *N* hydrochloric acid solution to a pink end point. Calculate the percent sodium oxide as follows:

$$\text{Percent sodium oxide} = \frac{3.1\ VN}{W}$$

Where:

V = ml. of hydrochloric acid solution used.
N = normality of hydrochloric acid solution used.
W = weight of sodium silicate solution in the aliquot, in grams.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.

5.1.1 *Levels A, B, and C.* One gallon, five, or fifty-five gallons as specified (see 6.2), of sodium silicate solution of one class (see 1.2), shall be packaged and packed, for each level of protection in appropriate containers in accordance with type II, class 1 or 2, of PPP-C-300.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 *Class* 1. Class 1 material covered by this specification is intended for use as an adhesive (in sealing paper cartons of corrugated or plain fiber and in affixing paper labels to fiber, paper, and wood surfaces), as a surface hardener for cement, as a coating for asbestos pipe insulation, and for similar purposes.

6.1.2 *Class* 2. Class 2 material covered by this specification is intended for use as an adherent and as a coating for asbestos cloth pipe lagging and for similar purposes.

6.1.3 *Class* 3. Class 3 material covered by this specification is intended for use as a surface coating for impression rubbers on intaglio printing presses to prevent listing of printed sheets.

6.1.4 The sodium silicate solutions covered by this specification may be thinned with water if desired. Residual sodium silicate should be washed thoroughly with water from laboratory glassware to avoid formation of an insoluble film.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and indicate the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Class of material required (see 1.2).
(*c*) Size container required (see 5.1).
(*d*) Selection of the applicable levels of packaging and packing (see 5.1).

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to those documents for guidance.

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MILITARY INTEREST:

CUSTODIANS:

Army—GL
Air Force—68

Review activities:

Army—GL, MD, MU
Air Force—68

Preparing activity:

Army—GL

CIVIL AGENCY INTEREST:

Interested activities:

AGR
COM
FPI
GSA
INT
JUS

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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